Classification of Hyperspectral Image Based on Task-Specific Learning Network

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***Abstract*— Hyperspectral image classification, which is a crucial task for various remote sensing applications, can achieve qualified performance under a conventionalized assumption, i.e., there are sufficient samples in every concerned class. However, in field investigation, collecting enough samples for every class is extremely difficult, which results in defective training sets with insufficient or imbalanced samples and deteriorates the classification performance dramatically. In order to address this issue, we propose a hyperspectral image classification method based on a task-specific learning network. The proposed network works under an episode-based framework, which learns general knowledge from the tasks with sufficient samples by metalearning and relation learning and then inference specific knowledge for the tasks with few samples by parameters adjustment and representation comparison. In particular, a task-specific feature learner is designed to learn unbiased features, and a comparison-based classifier is utilized to adapt the minority classes. As a result, the proposed method can obtain a qualified overall accuracy over all samples with a comparative averaged accuracy over all classes. Extensive experiments on three real hyperspectral images show that our method can achieve state-of-the-art performance under both the few-shot and imbalanced training set settings.**

***Index Terms*— Classification, hyperspectral image, metalearning.**

I. Introduction

SEMANTIC classification of remote sensing images is fundamental and crucial for various observation applications, such as land-use investigation, vegetation recognition, and environment monitoring. Hyperspectral images captured on remote sensing platform, which can provide high-resolution spectral information and large-scale spatial information about the scene of interest, are extensively used as one of the most important data sources in the classification tasks of remote sensing images [1], [2]. Therefore, hyperspectral image classification is an important research topic and application task in remote sensing area [3]–[5]. However, because of the large data volume caused by the high spectral resolution and extensive spatial coverage, and the mixed pixels prompted by the diffuse reflection of light and the relatively low spatial resolution, hyperspectral image classification is still a valuable yet challenging research problem and has attracted a lot of attention [6].

During the past decade, instead of traditional visual interpretation methods, most hyperspectral image classification methods are developed based on a machine learning theory [7]. Multiple kernel learning [8], low-rank representation [9], and joint sparse representation [10] are all successfully used in hyperspectral image classification and achieve the state-of-the-art performance. In particular, deep learning-based methods, which utilize deep and hierarchical neural networks to learn abstract and representative features, become a new development trend in hyperspectral image classification [11]–[16]. However, the aforementioned hyperspectral image classification methods can achieve qualified performance under a conventionalized assumption that there are sufficient training samples in every concerned class, i.e., the training samples of every class should be enough for learning general knowledge of the concerned class. Due to the difficulties in field investigation and the uneven distribution of land-cover types, it is hard to collect sufficient labeled training samples for every concerned class, which results in imbalanced samples between different classes. The aforementioned issue leads the classification model to emphasize knowledge on the majority classes, which may produce classification results with tolerable overall accuracy but unqualified averaged accuracy. Especially, when the training samples are insufficient for every class, the classification model works under a few-shot setting, and the overall accuracy also drops dramatically.

In order to address the imbalanced classification problem, some works refer to sample generation or reorganization methods, Li *et al.* [17] utilized orthogonal complement subspace projection (OCSP) to select samples from majority classes and created artificial samples for minority classes to balance different classes, Zhu *et al.* [18] used adversarial learning

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to generate fake samples and improved the generalization capability of the discriminative convolutional neural network, Sun *et al.* [19] partitioned the majority classes into different groups with small sizes to realize class balance. Moreover, some works focus on the classification with insufficient samples and propose few-shot classification methods for hyperspectral images. Pan *et al.* [20] proposed a multigrained network to use limited samples for hyperspectral image classification, Zheng *et al.* [21] proposed a method based on superpixel segmentation and distance-weighted linear regression classifier to tackle the small training set classification problem, Zhao *et al.* [22] proposed a multiple-feature learning-based model with adaptive weights for classification with limited training samples, and we utilize metalearning and relation learning to borrow the knowledge of other hyperspectral images [23]. When challenged by a defective training set, all the aforementioned methods focus on the specific situations of the classification task and achieve better performance than the traditional ones. However, the heavy dependence on samples quantity and distribution still limits the extensive applications of hyperspectral image classification methods. In this article, we try to improve the performance when only defective training sets are available and propose a hyperspectral image classification method based on a task-specific learning network (TSL-Net) to classify with imbalanced or insufficient samples.

In particular, we propose a TSL-Net to achieve satisfying classification performance under imbalanced or insufficient training setting. The proposed method includes a task-specific feature representation module that adjusts feature learning parameters according to the task representation and a comparison-based classifier that classifies the features by maximizing the representation similarity. The two modules work under an episode-based framework to adapt the imbalanced and insufficient classification problem and obtain a qualified overall accuracy over all samples with a comparative average accuracy over all classes. The major contributions can be summarized as follows.

1) We design a task-specific feature learning method to adapt different classification tasks, which generates the class representation from the sample representation and deduces the task representation from the class representation and hereby learns task-specific parameters to update the sample representation. The feature learner adjusts the parameters with the task knowledge, which can learn unbiased features for the minority classes.
2) We propose a comparison-based classification strategy to classify using deficient training sets, which learns to compare different representations with multiple fully connected layers and classifies the testing samples with the maximized similarity. The classification strategy learns the sample relation other than the class distribution, which can be improved by metalearning on other data sets hereby more suitable for insufficient samples.
3) We utilize an episode-based framework to realize the proposed hyperspectral image classification method, which builds each episode with an example set and a query set and completes both the training and the testing by multiple episodes. The framework breaks the training set into small subsets, which reaches sample balance by equally random sampling.

II. Related Work

There are two major steps in the framework of hyperspectral image classification: feature extraction and classification. The former one focuses on extracting features from the input samples, and the latter one emphasizes classifying the extracted features. The two steps work together to produce qualified classification results. This section gives a brief introduction to feature representation and classification methods for hyperspectral images.

A. Feature Extraction Method

Feature extraction extracts discriminative and representative features from the input samples, which is an important step in all classification-related methods. Due to the high-dimensional spectral and spatial information, the hyperspectral image depends heavily on the feature representation methods to produce high accuracy results. During the past decade, many feature extraction methods have been proposed for hyperspectral image classification. Kang *et al.* [24] inspired by edge-preserving filters and proposed a principal component analysis-based edge-preserving filters method to extract features of hyperspectral image, Luo *et al.* [25] studied on discriminant analysis and proposed a spatial–spectral hypergraph discriminant analysis method to learn spatial and spectral information, and Xia *et al.* [26] combined random forest ensembles and extended multiextinction profiles and designed a hyperspectral image feature extraction method. However, the aforementioned methods extract features based on prior knowledge about the input data set and cannot learn adaptive and abstract features for the classification tasks.

Recently, due to the advantages of abstract feature learning, feature extraction methods based on deep learning are widely used for feature extraction of hyperspectral images and produce the state-of-the-art performance. According to the particularity of hyperspectral images, some effective deep networks have been proposed for feature extraction of hyperspectral images, Paoletti *et al.* [27] proposed a 3-D convolutional neural network to effectively learn both spatial and spectral features, Chen *et al.* [28] designed a deep ensemble learning framework and successfully combined ensemble learning with deep network, Sidike *et al.* [29] introduced a progressively expanded neural network according to the specificity of hyperspectral image, and Zhu *et al.* [18] developed a generative adversarial network for hyperspectral image classification. With the excellent feature learning ability, deep learning has improved hyperspectral image classification to a new level. The excellent performance of deep networks relies on the tons of network parameters, which require lots of samples to learn. However, using imbalanced and insufficient training samples, it is hard for deep networks to learn accurate and unbiased parameters.

In this article, we train a deep network using an episode-based framework with equally random sampling to balance training samples from different classes and design a

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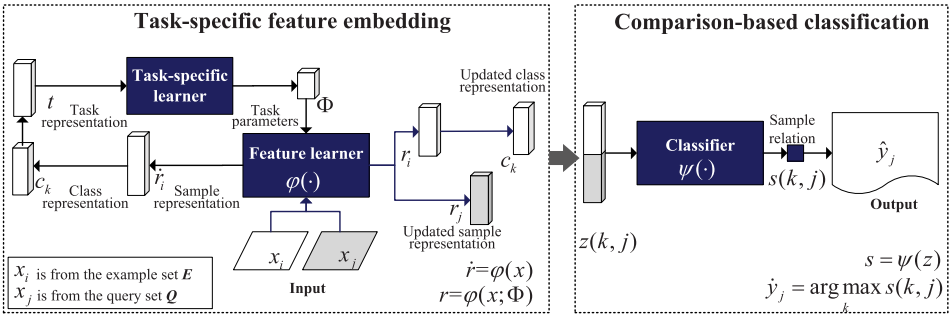


Fig. 1. Proposed TSL-Net. Two modules are embedded to learn unbiased information from both majority classes and minority classes, including a task-specific feature embedding module for extracting features and a comparison-based classification module for producing a classification map.

special structure to learn features according to the task situation. To this end, the proposed network can learn task-specific parameters and extract unbiased features for all classes.

B. Classification Method

Classification method is the key step of hyperspectral image classification, which decides the classification performance directly. During the past decade, classifiers evolve from the traditional visual interpretation to machine learning-based methods. Supervised classification methods, which learn knowledge from a labeled training set, can achieve the best performance. Peng *et al.* [30] designed a joint sparse representation model with maximum likelihood estimation for supervised classification of hyperspectral images. Tu *et al.* [10] combined the correlation coefficient and joint sparse representation to use the within-class similarity between training and test samples while decreasing the between-class interference. However, when the labeled training samples are not enough to learn a set of generalized classification parameters, the performance of the aforementioned methods drops dramatically.

In order to mitigate the aforementioned problem, some works refer to unlabeled samples, Liu *et al.* [31] proposed a semisupervised hyperspectral image classification method based on the Laplacian least-squares support vector machine. Some others take the advantage of ensemble learning, and Feng *et al.* [32], [33] proposed adaptive ensemble methods based on a synthetic minority oversampling technique. When the training set contains enough samples to learn the distribution or general knowledge of each class, all the aforementioned methods achieve great performance. However, it is difficult to collect sufficient samples for every class in remote sensing applications.

When the training set is not balanced, i.e., some of the classes do not have enough training samples, we think that it is better to refer to classification methods based on metric learning. These methods do not depend on learning the distribution of each class and can be pretrained on other data sets, which performs classification by a comparison-based strategy. Therefore, we proposed a comparison-based classification method, which classifies every test sample by the relation between a known sample and the test samples.

III. Methodology

In this article, we propose a hyperspectral image classification method that tries to produce qualified performance when the labeled training samples are insufficient in some classes or imbalanced between different classes. As shown in Fig. 1, the proposed method is realized under an episode-based framework to balance the samples between different classes, which completes the training and the testing of a TSL-Net by multiple randomly built subsets. Especially, two modules are embedded in the framework to equally learn information from both majority classes and minority classes, which is a task-specific feature learning module for extracting features and a comparison-based classifier for producing classification map. With the episode-based framework and the comparison-based classier, the whole network can be pretrained on other data sets by metalearning. The detailed information of the framework and the two modules will be presented in Sections III-A–III-C.

For a clear presentation, we briefly present the problem statement and the symbols explanation. Suppose that $\boldsymbol{x}$ is the input sample and y is the corresponding label. $\boldsymbol{x}$ is a vector of spectral information at a pixel of the hyperspectral image or a 3-D matrix of spectral information within a small patch of the hyperspectral image. We also use $\boldsymbol{y}$ to denote the vectorized label, which is a vector with the location of 1 to indicate the sample's class information. We define the training set as $\boldsymbol{X} = \{\boldsymbol{x}_i\}_{i=1}^{N}$ and the corresponding label set as $\boldsymbol{Y} = \{y_i\}_{i=1}^{N}$, where N is the number of samples in the training set. The sample label $y_i \in \{1, 2, \ldots, K\}$, where K is the number of classes. Suppose that the sample number of the kth class is n_k, where $N = \sum_{k=1}^{K} n_k$. For succinct expression, we refer to the class with sufficient samples as the majority class and the class with few samples as the minority class. If all the classes are minority classes, i.e., n_k is a small number for $k = 1, \ldots, K$, which indicates that there are not enough labeled samples for training, the problem becomes a few-shot classification problem. If n_k ranges from a small number to a large number for $k = 1, \ldots, K$, which means that the accuracy of the majority classes will shadow the minority classes, the problem becomes an imbalanced classification problem. In this section, we focus on the aforementioned problems and try to achieve qualified overall accuracy and averaged accuracy under both training settings.

A. Task-Specific Feature Embedding

Usually, the feature embedding module can learn the representations of all samples with a uniform set of parameters, i.e., once the training process is finished, the parameters

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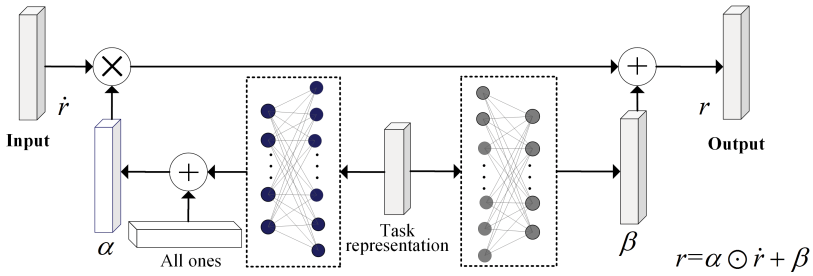


Fig. 2. Task-specific feature learning module.

are constant for all samples. However, there are not enough samples to learn general knowledge for the minority class, i.e., the parameters for classifying the minority class are not robust. Moreover, when encountered with the majority classes, the samples of the minority class will be treated as noise and shadowed by the majority classes. To deal with this problem, we design a task-specific feature learning module that generates a set of task-specific parameters by a subnetwork and adjusts all feature embedding parameters according to the task-specific parameters.

The embedding function of the original feature embedding module is defined as φ, and we train a subnetwork to learn a set of parameters Φ for each task and then update the feature embedding function of the backbone network with the learned task-specific parameters as $\varphi(\boldsymbol{x}; \Phi)$. In the beginning, we define three representations. The sample representation is defined as the feature of the input sample, which is the output of the feature embedding module. The class representation is defined as the feature of the class within one task. The task representation is defined as the feature of the task, which is consisted of multiple classes with several samples. Mathematically, in order to learn the task representation and give a specialized knowledge of the task, we compute the class representation first. The class representation is defined as the mean value of the sample representation that belongs to the same class within the task

$$\boldsymbol{c}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} 1\{y_i = k\} \dot{\boldsymbol{r}}_i \tag{1}$$

where $\dot{\boldsymbol{r}}_i$ is the sample representation of the original feature embedding module, n_k is the number of samples that belong to class k, and $1\{y_i = k\}$ is the indicator function, which returns to 1 if the equation is true and 0 otherwise. Moreover, we define the task representation as the mean value of the class representation within the task

$$\boldsymbol{t} = \frac{1}{K} \sum_{j=1}^{K} \boldsymbol{c}_k. \tag{2}$$

We utilize the task representation as the input of the task-specific subnetwork and learn two parameters: the scale parameter $\boldsymbol{\alpha}$ and the bias parameter $\boldsymbol{\beta}$. The embedding network is shown in Fig. 2, which is consisted with two branches to learn two parameters. Both branches utilize multiple fully connected layers to learn the task-specific parameters, and the output of the last layer has the same dimension as the task representation, which is also equal to the dimension of the sample representation. Moreover, for the scale parameter, an all-one vector is added to the output of the last layer to make sure that the updated features share the same amplitude coverage. The updated task-specific representation is computed by

$$\boldsymbol{r}_i = \boldsymbol{\alpha} \odot \dot{\boldsymbol{r}}_i + \boldsymbol{\beta} \tag{3}$$

where $\odot$ means the elementwise multiply and $\dot{\boldsymbol{r}}_i$ is the output of the original feature embedding module.

B. Comparison-Based Classification

Through the feature embedding module, we can get the spectral–spatial features for all samples from both the example set and the query set. Suppose that the example set is denoted as $E = \{\boldsymbol{x}_i\}_{i=1}^{N_E}$, and the query set is denoted as $Q = \{\boldsymbol{x}_j\}_{j=1}^{N_Q}$. We refer to each sample in E as an example and each sample in Q as a query sample. Samples $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ are from the example set and the query set, respectively, and $\boldsymbol{r}_i = \varphi(\boldsymbol{x}_i; \Phi)$ and $\boldsymbol{r}_j = \varphi(\boldsymbol{x}_j; \Phi)$ are the corresponding features, i.e., the output of the feature embedding module. From the updated sample representations, we can update the class representation by 1. We concatenate the representation of each class with the representation of each query sample, $\boldsymbol{z}(k, j) = (\boldsymbol{c}_k; \boldsymbol{r}_j)$, which is taken as the input of the comparison-based classification module. This module consists of multiple fully connected layers, and each layer takes the output of the previous layer as input and projects it into a new space by multiple network parameters and an active function. The former layers play the role of information optimization, and the last layer projects the feature into a relation score, which indicates the similarity between the query sample and the example. Suppose that the embedding function of the relation learning module is denoted as ψ, and then, the relation score is $s(k, j)$ is computed by

$$s(k, j) = \psi(\boldsymbol{z}(k, j)). \tag{4}$$

We compute the relation score between the sample representation and the class representation and project $\boldsymbol{z}(k, j)$ into a relation score, which indicates the similarity between the query sample and a class, and the label will be decided by the maximized relation score by (5). The query sample should be a member of the most related class, which has the maximum of class relation scores.

C. Episode-Based Framework

The episode-based framework performs both the training process and the testing step with multiple episodes, which are randomly built subsets from the training set and the testing set. Each episode consists of an example set and a query set. Suppose that the example set is denoted as $E = \{\boldsymbol{x}_i\}_{i=1}^{N_E}$

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and the query set is denoted as $Q = \{\boldsymbol{x}_j\}_{j=1}^{N_Q}$. We refer to each sample in E as an example and each sample in Q as a query sample. During training, each example set and query set are built by randomly selected samples from the training set, and the proposed network learns to represent all samples, i.e., both examples and query samples, and then learns to compare the query samples and examples in feature space. During testing, each example set is also constructed by randomly selected samples from the training set, but each query set is built by randomly selected samples from the testing set, and the proposed network represents all samples, i.e., both examples and query samples, and classifies all the query samples. Mathematically, taking training as an example, we complete an episode-based framework by repeating the following three major steps.

1) *Building an Episode:* We select M_1 labeled samples from each of the K classes to build an example set $E = \{\boldsymbol{x}_i\}_{i=1}^{N_E}$, where $N_E = M_1 \times K$, and select M_2 labeled samples from each of the K classes to construct a query set $Q = \{\boldsymbol{x}_j\}_{j=1}^{N_Q}$, where $N_Q = M_2 \times K$.
2) *Learning to Represent:* We feed each example $\boldsymbol{x}_i$ and each query sample $\boldsymbol{x}_j$ into the feature embedding module to produce spectral–spatial features. The embedding function is denoted as φ with a set of task-specific parameters Φ, and the spectral–spatial features are $\boldsymbol{r}_i = \varphi(\boldsymbol{x}_i; \Phi)$ and $\boldsymbol{r}_j = \varphi(\boldsymbol{x}_j; \Phi)$, respectively. We compute the class representation $\boldsymbol{c}_k$ by the sample representations of the example set.
3) *Learning to Compare:* We concatenate the representation of each query sample with the representation of each class $\boldsymbol{z}(k, j) = (\boldsymbol{c}_k; \boldsymbol{r}_j)$ and feed $\boldsymbol{z}(k, j)$ into the classification module to compare features. The embedding function is denoted as ψ, the relation score is $s(k, j) = \psi(\boldsymbol{z}(k, j))$, for $k = 1, \ldots, K$, and the label of the query sample $\hat{y}_j$ is determined by the example given the highest relation score

$$\hat{y}_j = \arg\max_k s(k, j). \tag{5}$$

During the training process, we minimize the following objective function to train the whole episode-based framework:

$$L(\Theta) = \sum_{j=1}^{N_E} \|\hat{\boldsymbol{y}}_j - \boldsymbol{y}_j\|_2^2 \tag{6}$$

where Θ is the parameter vector of the whole network and $\hat{\boldsymbol{y}}_j$ is the vectorized label of $\hat{y}_j$. During testing, we randomly select M_1 training samples from each of the K classes to build an example set $E = \{\boldsymbol{x}_i\}_{i=1}^{N_E}$, where $N_E = M_1 \times K$, and select N_Q testing samples to construct a query set $Q = \{\boldsymbol{x}_j\}_{j=1}^{N_Q}$. The episode-based framework utilizes the parameters learned in the training process to represent the example and the query sample and then compare the query representation with the example representation to decide the label information.

The episode-based framework learns the ability to compare different samples, which makes the network learn transferable knowledge. Therefore, in order to promote the classification performance, metalearning on other data sets is utilized to pretrain the proposed network and then fine-train on the target data set to further adjust all pretrained parameters.

IV. Experiments

In this section, we analyze the performance of the proposed TSL-Net using three real hyperspectral images. Specially, we present the performance comparison with some related classification methods and also analyze all related parameters of the proposed TSL-Net. To be fair, all the experiments are implemented or reproduced with Python on the platform of a desk computer with Intel Core i7 4.0-GHz CPU, NVIDIA GeForce RTX 2080Ti GPU, and 32-GB memory.

A. Experimental Data Set

We evaluate the performance of the proposed TSL-Net using three real hyperspectral images, which are all widely used in the works of hyperspectral image classification and can be acquired from the website of Computational Intelligence Group from the Basque University[1].

1) Indian Pines Scene: The hyperspectral image was gathered by the Airborne Visible/Infrared Imaging Spectrometer over Northwest Indiana, USA, 1992, which is consisted of 200 bands with 145×145 pixels. The spatial resolution is about 20 m. The corresponding ground-truth map contains 16 labeled land-cover types, mainly are corps in different growth phases. These corps share similar spectral characters, thus making the classification of this data set more difficult. The pseudocolor image (composition of bands 50, 27, and 17) and the available ground-truth map with color index are shown in Fig. 3(a). For clear display, all unlabeled samples are indicated with white color in the ground-truth map.

2) University of Pavia Scene: The hyperspectral image was acquired by the Airborne Visible/Infrared Imaging Spectrometer over Pavia city, Northern Italy, 2002, which is consisted of 103 bands with 610×340 pixels after removing a black region of 610×210 pixels and 12 bands. The spatial resolution is about 1.3 m. The ground-truth map contains nine land-cover types, which includes buildings and city green belt. The pseudocolor image (composition of band 60, 30, and 2) and the ground-truth map with color index are shown in Fig. 3(b).

3) Salinas Scene: The hyperspectral image was captured by the Reflective Optics System Imaging Spectrometer over Salinas Valley, California, which is consisted of 204 bands with 512×217 pixels after discarding water absorption bands. The spatial resolution is about 3.7 m. The ground-truth map contains 16 land-cover types, such as vegetables, bare soils, and vineyard fields. The pseudocolor image (composition of band 50, 27, and 17) and the available ground-truth map with color index are shown in Fig. 3(c).

B. Experimental Setup

The proposed TSL-Net focuses on two issues of hyperspectral image classification, i.e., the few-shot classification aroused by insufficient samples and the imbalanced classification caused by imbalanced samples between classes. Therefore, two parts with different training setups are included in our

[1]All data sets used in this article are publicly available: http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

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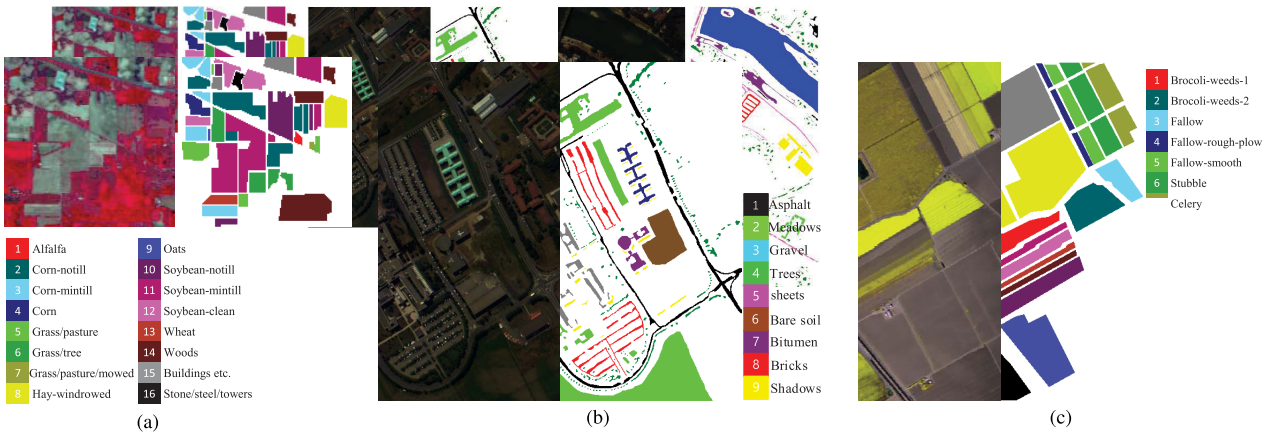


Fig. 3. All hyperspectral scenes used in this article, including the pseudocolor images and the corresponding ground truths with color indexes. (a) Indian Pines scene. (b) Pavia scenes. (c) Salinas scene.

TABLE I

NETWORK PARAMETERS FOR ALL THREE DATA SETS, INCLUDING THE INPUT SIZE, THE KERNEL SIZE, AND NUMBER OF ALL CONVOLUTIONAL LAYERS, THE POOLING SCALE AND STRIDE, AND THE OUTPUT SIZE

Dataset	input	conv.1	conv.2	pool.	conv.3	pool.	output
Indian Pines	17×17,200	1×1,180	3×3,240	2×2,2	3×3,300	5×5,1	300
University of Pavia	23×23,102	1×1,80	3×3,124	2×2,2	3×3,256	8×8,1	256
Salinas	21×21,200	1×1,180	3×3,240	2×2,2	3×3,300	7×7,1	300

experiments. Three hyperspectral image classification methods are utilized as comparing methods in both parts. The directly supervised method based on deep network (Sup-Net), which uses the feature embedding module in the proposed network with Softmax as a classifier, tries to explore the information of all labeled samples exhaustively. Sup-Net indicates the classification result with the insufficient or imbalanced labeled training samples only, and it serves as a comparison baseline. A semisupervised method based on deep network (Semi-Net), which also uses the feature embedding module in the proposed network with Softmax as a classifier, takes advantage of both labeled samples and unlabeled samples by a multi-decision strategy. Semi-Net gives a clue of the classification result with the target data set only. Hybrid spectral CNN (HybridSN) [34] based on CNN, which utilizes 3-D-CNN to learn a joint spatial–spectral feature representation, uses 2-D-CNN on top to further learn more abstract-level spatial representation. HybridSN indicates the performance of the CNN-based method. Moreover, for few-shot classification, a few-shot hyperspectral image classification method based on a relation learning network (RL-Net) is utilized as the state-of-the-art comparing method. RL-Net is a metalearning method that is transferred between different images for general information sharing and fine-trained on a specific hyperspectral image for individual information learning. For imbalanced classification, a method based on OCSP is used as the state-of-the-art comparing method. OCSP is a data augmentation method, which selects samples from large-size classes and create artificial samples for small-size ones to balance different classes. For a fair comparison, we generate balanced samples through the OCSP method and then use all available samples to train the feature extractor and classifier of the supervised method.

During experiments, we utilize a convolution neural network as the backbone structure of the TSL-Net, which is consisted of three convolutional layers and two pooling layers. The detailed configurations of the backbone network for all three hyperspectral images are shown in Table I, where conv. denotes the convolutional layer and pool. means the pooling layer. The second column indicates the space size of the input samples, for example, from the second row, the hyperspectral image of Indian Pines scene is cropped into 17×17 small patches over all 200 bands. The table shows the kernel size and number of all convolutional layers, such as the second convolutional layer of the Salinas scene that uses 240 kernels with spatial size 3×3. For pooling layers, such as $2 \times 2, 2$, it means that the size of pooling is 2×2, and the stride is also 2. The last column of the table indicates the output dimension of the features. During training, since the smallest numbers of training samples of all three data sets are 5, we set the M_1 to 2 and set the M_2 to 3.

In order to give an objective evaluation of the proposed TSL-Net, three widely used measurements are utilized for all experiments. Overall accuracy (OA, %) is the averaged accuracy over all testing sample, average accuracy (AA, %) is the averaged accuracy over all concerned classes, and Kappa coefficient (κ) is the robustness measurement with the degree of agreement. From the definitions of different evaluations, we can conclude that the low accuracy of a minority class will affect OA slightly but decrease AA observably. Moreover, we also provide the class accuracy (CA, %), which is averaged accuracy over one certain class.

C. Experimental Result

In this section, for both the few-shot classification and the imbalanced classification, we presented the classification maps

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TABLE II

CLASSIFICATION PERFORMANCE REPRESENTED BY CA(%), OA(%), AA(%), AND κ COEFFICIENT OF DIFFERENT METHODS FOR THE INDIAN PINES SCENE

Class	Training	Testing	Sup-Net	Semi-Net	HybridSN	RL-Net	TSL-Net
1	5	41	99.21	98.43	97.56	88.05	**100.00**
2	5	1423	56.64	**65.07**	23.96	36.75	54.39
3	5	825	36.36	61.48	39.75	59.52	**62.45**
4	5	232	88.36	92.87	74.14	96.98	**100.00**
5	5	478	49.17	73.01	74.06	**78.24**	70.20
6	5	725	63.45	96.83	94.90	97.52	**98.34**
7	5	23	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
8	5	473	76.11	86.28	**100.00**	99.58	**100.00**
9	5	15	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
10	5	967	71.77	75.74	64.22	78.39	**81.80**
11	5	2450	58.80	47.55	49.76	**59.31**	51.59
12	5	588	55.61	57.94	30.10	58.59	**68.89**
13	5	200	98.50	**100.00**	**100.00**	**100.00**	**100.00**
14	5	1260	**87.70**	80.35	79.69	75.32	86.75
15	5	381	74.28	**95.48**	39.11	94.75	82.73
16	5	88	96.42	**100.00**	76.14	**100.00**	**100.00**
OA			65.00	69.29	57.73	70.17	**70.97**
AA			75.67	81.67	71.46	82.56	**83.63**
$\kappa(\times 100)$			60.81	64.34	53.05	66.65	**67.62**

of the proposed TSL-Net and all the comparing methods and summarized the corresponding accuracy with related analysis.

1) Few-Shot Classification Experiment: We use the hyperspectral image of Indian Pines scene to evaluate the performance of few-shot classification. Concretely, we use the Salinas data set as the source data set to pretrain the proposed network and take the Indian Pines data set as the target data set. For the target data set, we randomly select five labeled samples from each class to build the training set, in which the number of training samples is very small and meets the few-shot setting. The classification maps of Sup-Net, Semi-Net, HybridSN, RL-Net, and the proposed TSL-Net are shown in Fig. 4 and the corresponding CA, OA, AA, and κ coefficient are summarized in Table II.

From the results in Fig. 4 and Table II, we can conclude that the proposed TSL-Net can produce the best classification results under the few-shot setting. Because five labeled samples are not enough to represent the entire category, Sup-Net produces the lower accuracy. Semi-Net slightly improves the accuracy of the information provided by the unlabeled samples. Since HybridSN composed of 3-D-CNN and 2-D-CNN has the largest number of parameters, when the labeled samples of each category are 5, HybridSN achieves the lowest classification accuracy. RL-Net and TSL-Net can give better results by borrowing information from a source data set. TSL-Net can learn more elaborate information according to the task representation by the task-specific learning and produce the best classification results with only five labeled samples.

Moreover, since the classification stability is very important in the insufficient situation, we execute ten random tests for all methods and report the mean and the standard deviation of OA, AA, and κ coefficient in Table III. In each test, we randomly

TABLE III

PERFORMANCE ANALYSIS OF INSUFFICIENT CLASSIFICATION

Method	OA	AA	$\kappa(\times 100)$
Sup-Net	65.10±9.13	75.80±8.54	60.79±9.62
Semi-Net	69.24±6.31	81.57±5.65	64.40±7.29
HybridSN	57.53±7.63	71.51±6.28	53.00±6.72
RL-Net	70.37±5.48	82.44±**2.57**	66.66±4.85
TSL-Net	**71.01±4.96**	**83.62**±3.01	**67.70±3.74**

TABLE IV

CLASSIFICATION PERFORMANCE REPRESENTED BY CA(%), OA(%), AA(%), AND κ COEFFICIENT OF DIFFERENT METHODS FOR THE UNIVERSITY OF PAVIA SCENE

Class	Training	Testing	Sup-Net	Semi-Net	HybridSN	OCSP	TSL-Net
1	133	6498	**99.23**	98.83	99.14	98.25	97.41
2	373	17716	**100.00**	99.78	**100.00**	**100.00**	99.82
3	11	2078	73.40	84.03	77.63	81.34	**89.49**
4	62	2910	99.80	99.05	99.25	99.92	**100.00**
5	7	1338	28.57	49.00	**56.38**	50.07	50.64
6	101	4777	88.11	86.61	87.25	87.75	**90.32**
7	7	1323	67.83	**79.30**	72.08	75.69	76.98
8	74	3497	85.32	82.64	82.74	**89.65**	82.16
9	5	942	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
OA			93.21	94.48	94.21	94.63	**95.15**
AA			82.47	86.21	86.05	86.74	**86.87**
$\kappa(\times 100)$			90.90	92.63	92.26	92.82	**93.55**

selected five samples from the Indian Pines data set as the training sets. From the table, we can see that the proposed method can produce the highest averaged OA and κ coefficient along with the smallest variance, underlining its improved robustness against a particular choice of training samples. Although RL-Net can provide the smallest variance of AA, the proposed method can produce higher AA.

2) Imbalanced Classification Experiment: We use the University of Pavia scene and Salinas scene to evaluate the performance of imbalanced classification. In order to create more imbalanced training conditions, for the University of Pavia data set, we randomly select 0.5% of labeled samples for class 3, 5, 7, and 9, and 2% of labeled samples for the rest categories to build the training set, and for Salinas data set, we randomly select 0.5% of labeled samples for class 1, 3–5, 11–14, and 16, and 2% of labeled samples for the rest categories to build the training set. The second and third column of Tables IV and V show the sample number of each class in both the training set and the testing set. We can see that the sample numbers of some categories are less than one-tenth of other categories, for example, categories 2 and 9 in the University of Pavia scene. For the University of Pavia scene, we first pretrain TSL-Net with the Pavia Center scene, which is captured by the same sensor over similar area, and then use the training set from the University of Pavia to fine-train the network. For the Salinas scene, we first pretrain TSL-Net with the Indian Pines scene and then use the training set from Salinas scene to fine-train the network. The classification

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Fig. 4. Classification maps of different methods for the Indian Pines scene. (a) Sup-Net. (b) Semi-Net. (c) HybridSN. (d) RL-Net. (e) TSL-Net.

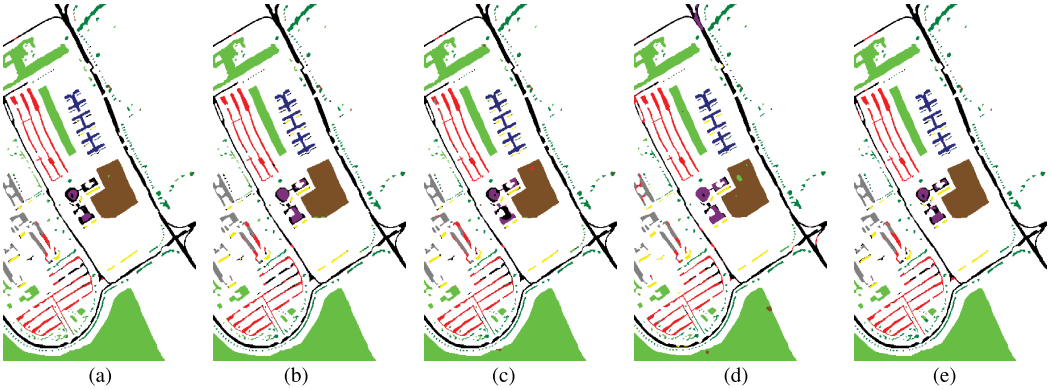


Fig. 5. Classification maps of different methods for the University of Pavia scene. (a) Sup-Net. (b) Semi-Net. (c) HybridSN. (d) OCSP. (e) TSL-Net.

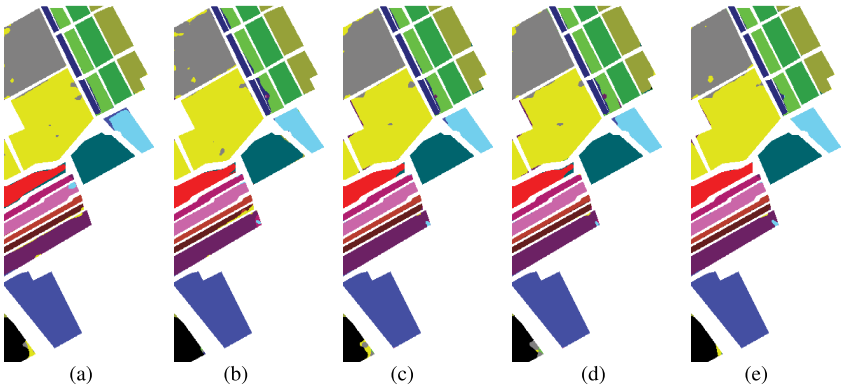


Fig. 6. Classification maps of different methods for the Salinas scene. (a) Sup-Net. (b) Semi-Net. (c) HybridSN. (d) OCSP. (e) TSL-Net.

maps of Sup-Net, Semi-Net, HybridSN, OCSP, and TSL-Net are shown in Figs. 5 and 6. Moreover, the corresponding averaged CA, OA, AA, and κ over ten trials are summarized in Tables IV and V.

From the classification maps and the corresponding accuracy, we can see that the proposed TSL-Net can produce the best performance for both scenes under imbalanced training setting. Sup-Net trains the deep networks using the imbalanced samples, which is easy to result in biased parameters for the minority classes. For example, in Table IV, the CA of categories 5, 7, and 8 is very low using Sup-Net. Semi-Net utilizes the information of unlabeled samples for information compensation, which can improve the accuracy slightly. HybridSN can extract enough spatial features and spectral features through 3-D-CNN and 2-D-CNN so that hyperspectral images can obtain a joint representation that is easier to classify. When training samples are enough, HybridSN can achieve better classification accuracy. OCSP creates new samples for minority classes by mapping the original data into orthogonal complement subspace, which balances the samples in different categories, thereby improving the accuracy to a certain extent. The proposed TSL-Net, which utilizes a task-specific feature learner to adapt imbalanced classes and designs a comparison-based classifier to treat all testing samples evenly, can produce the best OA, with the best CA and AA.

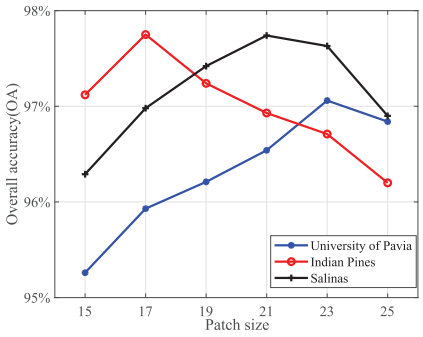


Fig. 7. Relation between patch size and classification performance.

D. Parameter Analysis

In this section, we analyze the related parameters of the proposed TSL-Net. In order to learn both the spatial and spectral information, we utilize data cube, including the spectral information of both the pixel under processing and

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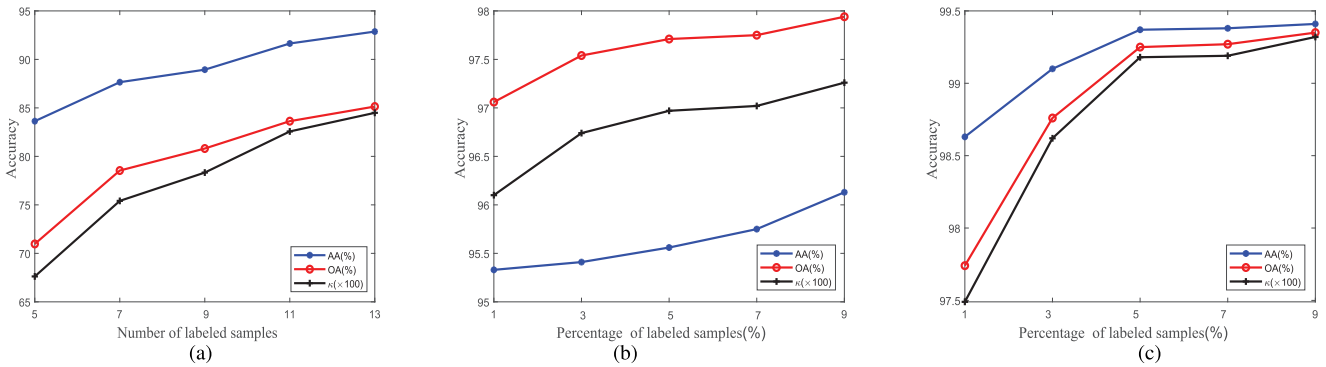


Fig. 8. Performance analysis over different numbers of training samples. (a) Indian Pines. (b) University of Pavia. (c) Salinas.

TABLE V

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS REPRESENTED BY CA(%), OA(%), AA(%), AND κ COEFFICIENT FOR THE SALINAS SCENE

Class	Training	Testing	Sup-Net	Semi-Net	HybridSN	OCSP	TSL-Net
1	11	1998	86.84	96.70	96.89	94.55	**99.69**
2	75	3651	**100.00**	98.66	**100.00**	**100.00**	**100.00**
3	10	1966	90.69	99.54	98.52	97.81	**99.89**
4	7	1387	98.60	**99.48**	99.12	98.75	98.68
5	14	2664	94.84	86.27	90.66	91.67	**96.67**
6	80	3879	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
7	72	3507	99.14	**99.94**	98.89	98.32	99.42
8	226	11045	98.08	**98.79**	97.69	97.04	98.06
9	125	6078	**100.00**	**100.00**	99.96	99.97	**100.00**
10	66	3212	95.98	96.01	**97.85**	97.76	97.07
11	6	1062	94.95	97.93	99.24	**99.52**	97.36
12	10	1917	95.20	98.59	99.37	**99.53**	99.26
13	5	911	96.74	96.89	**98.35**	92.42	96.04
14	6	1064	95.71	83.66	98.87	98.68	**98.96**
15	146	7122	98.17	95.94	97.30	**98.41**	98.20
16	10	1797	89.04	**97.50**	87.15	89.10	95.56
OA			97.28	97.53	97.70	97.64	**97.92**
AA			96.27	96.81	97.49	97.10	**98.37**
$\kappa(\times 100)$			96.96	97.25	**97.39**	97.24	97.31

TABLE VI

CLASSIFICATION ACCURACY REPRESENTED BY OA(%), AA(%), AND κ COEFFICIENT OF WHETHER TO USE METALEARNING

Dataset	Meta-learning	Fine-training	OA	AA	$\kappa(\times 100)$
Indian Pines	×	✓	76.42	84.59	71.25
	✓	×	76.87	84.72	72.03
	✓	✓	**78.52**	**86.73**	**73.31**
University of Pavia	×	✓	96.35	94.52	95.17
	✓	×	81.60	83.32	76.30
	✓	✓	**97.06**	**95.33**	**96.10**
Salinas	×	✓	97.20	97.88	96.88
	✓	×	82.20	87.10	80.29
	✓	✓	**97.74**	**98.63**	**97.49**

the pixels from the neighboring area as input samples. For implementation, we crop patches surrounding the pixel under processing over all bands from the original hyperspectral image. The spatial size of the patch is an important parameter, which has an obvious effect on the performance of TSL-Net. The relation curve between the patch size and the classification performance for all three hyperspectral images is shown in Fig. 7. From the figure, we can see that the best patch size is 17 for Indian Pines scene, 23 for the University of Pavia scene, and 21 for the Salinas scene. We conclude that this parameter is closely related to the spatial resolution of the hyperspectral image, and the optimal patch size of the hyperspectral image with high spatial resolution is small; otherwise, the best patch size is large.

Moreover, we investigate the effect of metalearning and fine-training. The metalearning is the strategy to use the information from other data sets, and the fine-training is the strategy to use more related information from the target data set. For the University of Pavia data set, the Pavia Center data set is used as pretraining data set, and the Indian Pines data set and Salinas data set are used as pretraining data set for each other. The result is shown in Table VI, where the checkmark indicates that the related step is used. From the table, we can see that the metalearning, which takes the advantage of the knowledge from other data sets, can improve the classification accuracy, and the improvement is obvious when the labeled samples of the target data set are not enough. Therefore, using both metalearning and fine-training, we can achieve the highest accuracy.

Furthermore, we analyze how the number of training samples affects the performance of the proposed method. All the parameters are fixed except for the numbers of training samples per class. Classification accuracy plots under various training sizes are shown in Fig. 8. For the Indian Pines data set of insufficient classification, the training sets are set up with 5, 7, 9, 11, and 13 samples from each class, and for the other data sets of imbalanced classification, the training sets are set up with 1%, 3%, 5%, 7%, and 9% samples from each class. From the figure, we can tell that the OA monotonically increases as the size of training set. When the training samples are enough, all the evaluations tend to be stable.

Finally, we also present the running time of the proposed method in Table VII, where the training time is the time cost of both metalearning and fine-training. From the table, we can

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TABLE VII
RUNNING TIME (S) OF THE PROPOSED METHOD

Dataset	Training time	Testing time
Indian Pines	189	9
University of Pavia	161	44
Salinas	207	69

see that the training can be finished very quickly, which is acceptable for a lot of applications.

V. CONCLUSION

This article proposes a TSL-Net to classify hyperspectral images when there are only defective training sets. The proposed method works under an episode-based framework to balance training samples from different classes and designs a task-specific feature learning to learn unbiased features for the minority classes. Moreover, since we utilize a comparison-based classifier, the whole classification framework can be pretrained on other data sets. All the aforementioned strategies work together to improve the classification performance. Experimental results demonstrate the effectiveness of the proposed method when dealing with defective training sets.

The proposed method utilizes the mean of all intraclass features as the class representation, which is a simple and fast solution. However, some adaptive methods can produce more robust class representation, which will be studied next. Moreover, the proposed method takes a simple convolutional network for insufficient classification, and some advanced architectures, such as residual connection, can further improve the proposed method, which will be studied further.

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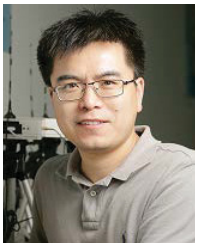
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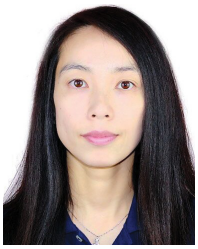
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